



A Toy Model of the Big Bang in 1+1 Dimensions with a ϕ^4 Field

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I, Francesca Darkins, confirm that I have read, understood, and have adhered to the plagiarism declaration above.

AI Usage Declaration

- “AI Overview that automatically appears at the beginning of every Google search was sometimes consulted”

Link to GitHub Codebase

All codes used to produce the analysis and results in this report can be found at the GitHub repository [here](#).

1 Introduction

Since kinks can be analogous to particles, I wanted to look into the production and behaviour of them in a semi-realistic scenario under random initial conditions. To do so, I created a toy model of the Big Bang in a 1+1 dimensional¹ world that contains only one field which will be modelled using the ϕ^4 model. I then simulated the results using finite difference schemes.

This document mainly focuses on describing the model itself and its implementation. The details of the results and observations from the simulations are discussed in more detail in the video.

2 Model Outline

The Big Bang started off with an incredibly hot and dense point that rapidly expanded. This expansion of space led to the energy being less dense and thus the temperature of space cooled. To mimic this behaviour for a fixed amount of space, I first put my initial conditions through the heat equation of the ϕ^4 model (1), then through the wave equation (2) to see how the model behaves.

$$\text{Heat equation} \quad \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2} - \frac{dU}{d\phi} \quad (1) \quad \text{Wave equation} \quad \frac{\partial^2 \phi}{\partial t^2} = \frac{\partial^2 \phi}{\partial x^2} - \frac{dU}{d\phi} \quad (2)$$

The initial conditions of the field are a set of uniformly distributed random points $\phi(t=0) \sim \mathcal{U}(-r, r)$ where r is a chosen boundary.

3 Model Implementation

The model was split into two stages: the heat equation and the wave equation, both modelled using finite difference schemes. In both cases, the spacial dimension was modelled using a central difference scheme with ghost nodes at each end taking on the value of the boundary node of the simulated region:²

$$\frac{\partial^2 \phi}{\partial x^2} \approx \frac{\phi_{x+1}^t - 2\phi_x^t + \phi_{x-1}^t}{\Delta x^2}$$

For the heat equation, a forward difference scheme was used to model the time dimension. As the initial conditions were random but not based off of any realistic phenomena, we only have information for $t=0$ and cannot analytically calculate the next time step. For the wave equation, a central difference scheme was used to model the time dimension. The initial conditions were put through the heat equation until $t=10+\Delta t_w$ where Δt_w is the time step used for the wave equation.³ Thus the solutions for the heat equation at $t=10$ and $t=10+\Delta t_w$ were the initial conditions used for the central difference scheme

¹1 spatial dimension + 1 temporal dimension

²The subscript and superscript notation used for finite difference schemes is the same as that used in the group project.

³Similarly Δt_h is the time step used for the heat equation.

for the wave equation. This led to the following equations for the numerical schemes modelling the heat equation (1) and the wave equation (2):

$$\begin{aligned} \text{Heat equation: } \phi_x^{t+1} &= \left(\frac{\phi_{x+1}^t - 2\phi_x^t + \phi_{x-1}^t}{\Delta x^2} + 2\phi_x^t(1 - (\phi_x^t)^2) \right) \Delta t_h + \phi_x^t \\ \text{Wave equation: } \phi_x^{t+1} &= \left(\frac{\phi_{x+1}^t - 2\phi_x^t + \phi_{x-1}^t}{\Delta x^2} + 2\phi_x^t(1 - (\phi_x^t)^2) \right) \Delta t_w^2 + 2\phi_x^t - \phi_x^{t-1} \end{aligned}$$

The finite difference schemes divide the simulated region into a grid of nodes. You can thus implement the random initial conditions by looking at each node of the grid x_i and assigning a value to the field ϕ at that point using the desired distribution of values. In our case, $\phi_{x_i}^{t=0} \sim \mathcal{U}(-r, r)$.

I modelled the region $x \in [-10, 10]$. This choice was arbitrary so I kept it consistent with the group project. I used $\Delta x = 10^{-2}$ which is the same as the group project, $\Delta t_h = 10^{-5}$ which satisfies the stability condition for the heat equation $\Delta t_h \leq \frac{1}{2}(\Delta x)^2$, and $\Delta t_w = 10^{-3}$ which satisfies the stability condition for the wave equation $\Delta t_w \leq \Delta x$. The boundary conditions used and justifications for choosing them are the same as that in the group project.

4 Some Kinks

This model has a flurry of interesting behaviours which I go through in the video. However just to show that kinks do indeed form in this model, here are some plots showing the model's behaviour in the wave equation using MATLAB's seed 6 and $r = 1$ for the initial conditions:

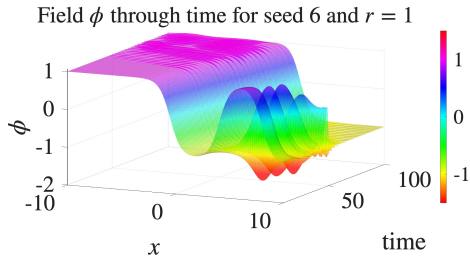


Figure 1: 3D plot of how the field ϕ changes through time

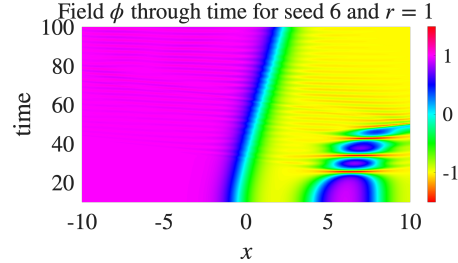


Figure 2: A colourmap showing how the field ϕ changes through time

I chose this colourmap because it clearly labels $\phi = -1$ as yellow, $\phi = 0$ as light blue, and $\phi = +1$ as purple. These values are important because they are the two vacuum states of the field and the halfway point between them, indicating which vacuum state a region is more likely to gravitate towards.

5 Flaws of the model

There are many variables in this model that have arbitrarily chosen values including what you bound your random initial conditions by (r) and how long you let your initial conditions 'settle' in the heat equation for.

Another issue is that if you run a simulation for long enough, you will see some highly oscillatory waves bombard the simulation from the boundaries. Because energy is conserved in the wave equation, even though the boundary conditions mean that the waves look as though they move out of the simulated region, the energy of those waves does not leave with them. The energy thus builds up at the boundaries until it manifests as highly oscillatory waves:

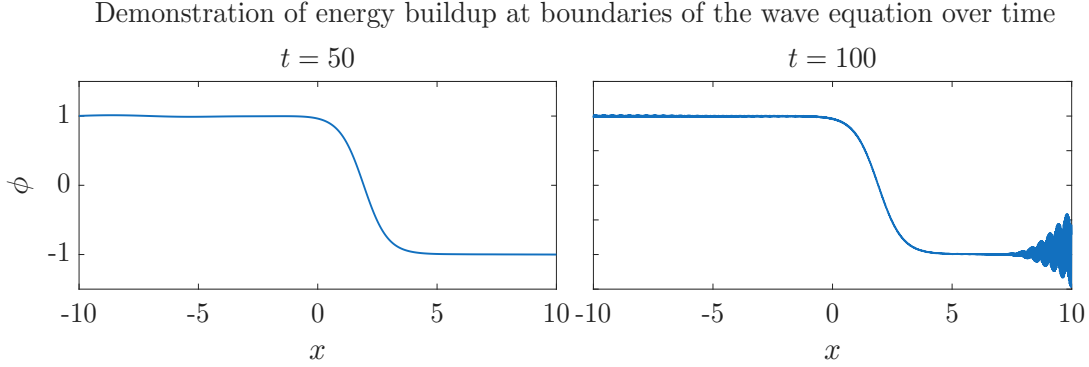


Figure 3: You can see the highly oscillatory wave start to emerge from the right boundary at $t = 100$. MATLAB's seed 4 and $r = 1$ were used for the random initial conditions in this simulation. The axis are the same on both graphs. The graphs aim to give a qualitative understanding of energy buildup so the y-axis on the right graphs has been omitted for visual simplicity.

A potential update of this model would be to use a combination of the heat and wave equation:

$$\frac{\partial^2 \phi}{\partial t^2} + \zeta \cdot \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2} - \frac{dU}{d\phi}$$

where ζ is the damping factor. In fact an even better model would have ζ be a decreasing function of time. As the universe never stopped expanding, an isolated space like our simulated region would still cool down locally. This rate of cooling would decrease as the universe expands. Thus a damping factor that decreases with time would mimic this phenomena, making the behaviour from the wave equation more dominant over time. The simplest forms of the damping factor to try first would be $\zeta(t) = (t - \frac{1}{\zeta_0})^{-1}$ or $\zeta(t) = \zeta_0 \cdot e^{-t}$ where $\zeta_0 := \zeta(0)$. Both have the benefit of decreasing with time without ever actually reaching 0 but decrease at different rates.

6 Conclusions

I have shown in my toy model that a 1+1 dimensional world consisting of one field of the ϕ^4 model starting with random initial conditions produces kinks. Since kinks can be analogous to particles, this also indicates particle production. If this approach was generalised to higher dimensions, used random initial conditions more synonymous to those of the Big Bang, and incorporated multiple interacting fields, it could shed light on how structures in our world formed in the early universe. Alternatively using a method going backwards in time starting with structures that we know exist now, we could learn about the initial conditions of the universe and investigate what the Big Bang looked like.